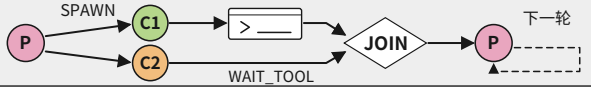
BeliefKV：工作流感知的执行与 KV 协同管理

运行时依赖 → 近期执行需求 → 页级迁移、驻留与原生准入

Agent harness：动态请求与完成信号



context	epoch	request
P / C	e	r

因果身份

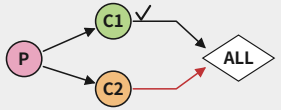
正文 / 通知

服务进度

tokens

BeliefKV 控制面

1 依赖状态 / RCCG



child	state	waiter
C1	RETURN	P
C2	NOTICE	P

GraphDelta　　pending = {C2}

2 异步执行需求预测



p_{final}

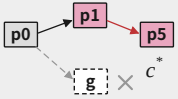
$[W_{\text{lo}}, W_{\text{hi}}]$

P JOIN　τ_{join}　τ_{tool}

模型预测需求；runtime 决定动作

3 Checkpoint、时机、容量与执行顺序

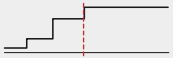
可复用 checkpoint



context / epoch / prefix ✓

ΔFULL + required state

实测传输启动窗口



$S_{90} + Q_{90}$

$\tau \le H$

容量与准入机会



aging / batch fill

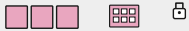
PREPARE_HOST



回收冷 FULL

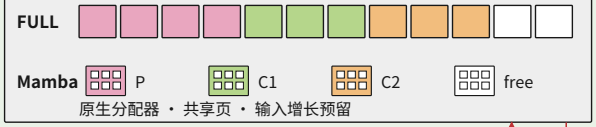


PREFETCH_GPU

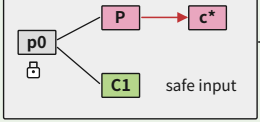


原生 SGLang 数据面

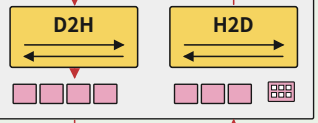
HBM: FULL pages + checkpoint state



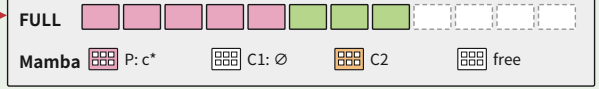
UnifiedRadixCache



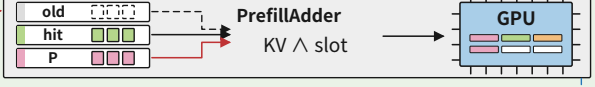
HiCache / DMA



Host 池：已备份 FULL 与有效状态



恢复感知的原生准入



Parent　Child 1　Child 2　Runtime 动作　ACK / service / first reuse